

HW/SW Co-design

Audio Spectrum Analyzer SoC

Presented to:

Analog Devices

Eng. Youssef k. Mohamed

Presented by:

Mohamed Abdelhafez Elnewehy

July 2026

1- SoC Architecture

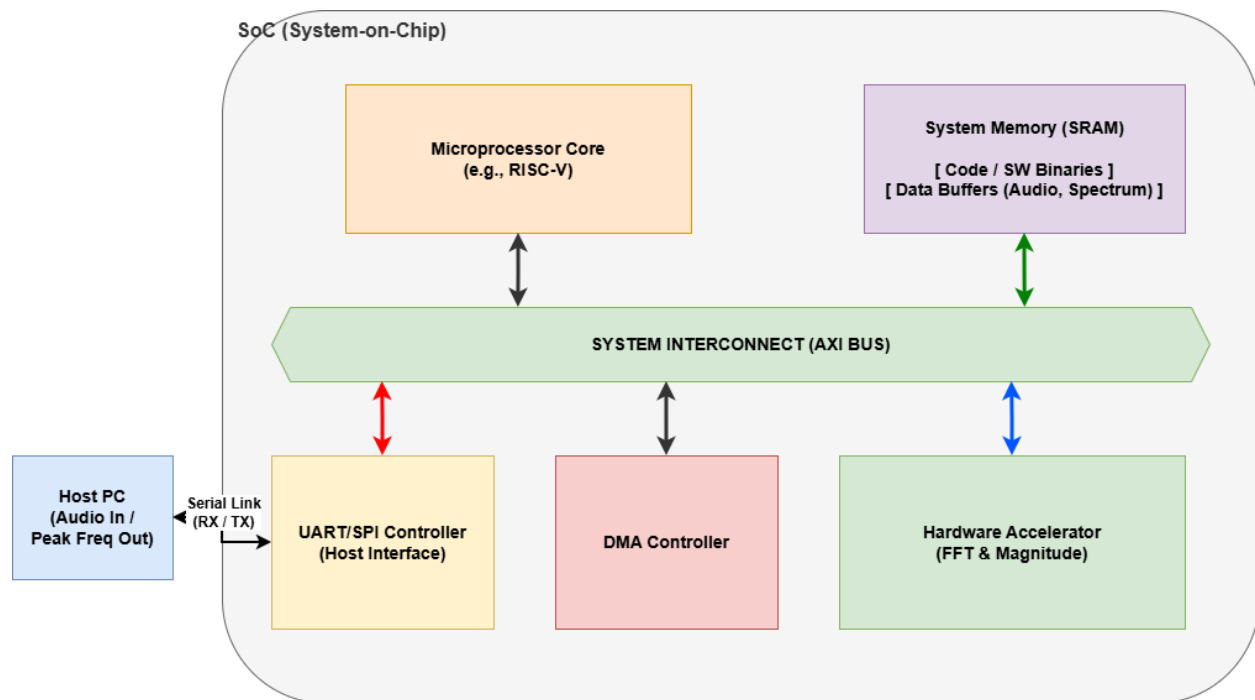


Figure 1 drow.io

2- Hardware/Software Partitioning with Reasoning

1. Store samples: Assigned to HW (DMA) to transfer incoming data quickly without pausing or interrupting the CPU.
2. Apply windowing: Assigned to SW because it is a simple sequential multiplication that the CPU handles easily without needing extra silicon area.
3. Compute FFT: Assigned to a HW Accelerator because the complex math $O(N \log N)$ is too heavy and slow to run in software.
4. Compute Magnitude: Assigned to the HW Accelerator to efficiently calculate square roots right after the FFT, saving time and memory transfers.
5. Find peak bin: Assigned to SW because searching an array for the highest number is a basic task perfectly suited for the CPU.
6. Convert to Frequency: Assigned to SW because it is just a single, simple arithmetic equation calculated once per block.

3- System Application Pseudocode

```
while (true) {  
  
    // 1. Receive Audio Data [HW: UART + DMA]  
    audio_samples = Receive_Audio_From_Host();  
  
    // 2. Apply Windowing [SW: Microprocessor]  
    for (int i = 0; i < N; i++) {  
        windowed_samples[i] = audio_samples[i] * window_coefficients[i];  
    }  
  
    // 3 & 4. Compute FFT & Magnitude [HW: Hardware Accelerator]  
    magnitude_spectrum = Run_Hardware_Accelerator(windowed_samples);  
  
    // 5. Find Peak Bin [SW: Microprocessor]  
    float max_value = 0;  
    int peak_bin = 0;  
  
    for (int i = 0; i < N/2; i++) {  
        if (magnitude_spectrum[i] > max_value) {  
            max_value = magnitude_spectrum[i];  
            peak_bin = i;  
        }  
    }  
  
    // 6. Convert Bin to Frequency [SW: Microprocessor]  
    float peak_frequency = peak_bin * (Sampling_Rate / N);  
  
    // 7. Transmit Result [HW: UART]  
    Send_Result_To_Host(peak_frequency);  
}
```